Stellar Metallicity from Synthetic PHOENIX Spectra: A Baseline PCA + Naive Bayes Pipeline and Its Limits

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Abstract

I built a small, reproducible pipeline that estimates stellar metallicity ([Fe/H]) from synthetic PHOENIX spectra using Principal Component Analysis (PCA) and a Gaussian Naive Bayes (GNB) classifier. On a 10×10 grid in $T_{\rm eff}$ and [Fe/H] (100 spectra), the classifier gets the exact metallicity bin right for only 20% of 25 test stars (88% within one 0.5 dex bin). Diagnostics show why: the first principal component carries 97.7% of the variance and tracks effective temperature, not metallicity. Leak-free cross-validation confirms the result (about 25% exact accuracy), and treating [Fe/H] as a continuous quantity gives an RMSE of 0.31 dex after a simple per-spectrum normalisation. I present this as an honest baseline and propose concrete steps towards Gaia/4MOST-like conditions.

1 Motivation

Gaia and 4MOST produce spectra for millions of stars, so fast automated parameter determination matters. Following our conversation at ISNA 2023, I built this small project to learn the full workflow (grid handling, preprocessing, dimensionality reduction, classification, validation) on a controlled synthetic problem before moving to realistic survey conditions.

2 Data and Method

Grid. PHOENIX spectra [1] were retrieved with the `expecto` package for 10 effective temperatures (4000–8500 K, step 500 K) and 10 metallicities (-3.0 to $+1.5$, step 0.5 dex) at fixed $\log g = 4.5$. The full grid of 100 spectra was recovered.

Preprocessing. Spectra were cut to 4000–7000 Å and reduced to 3,667 flux points by keeping every 100th sample (mean spacing 0.82 Å). This is simple decimation, not convolution to a defined resolution, so narrow lines are only sparsely sampled.

Model. Each flux column was standardised, compressed to 10 principal components, and passed to a GNB classifier with the 10 [Fe/H] values as classes. The original protocol used a random 75/25 split (seed 42, not stratified; 75 training and 25 test stars). The scaler and PCA were fitted on all 100 spectra before splitting, which is a small leak I correct below. Code uses scikit-learn [2]; all numbers in this report can be regenerated with this method GitHub Repository .

3 Results

3.1 16-star prototype

A 4×4 grid ($T_{\rm eff} = 5000$–6500 K, [Fe/H] $= -2, -1, 0, +0.5$) with 5 principal components gave 75% accuracy, that is 3 of 4 test stars. This showed the pipeline runs end-to-end but is too small to support any conclusion.

3.2 100-star grid

Following the successful prototype, the parameter space was expanded to a 10×10 grid, encompassing 10 effective temperatures ($4000\,\mathrm{K} \leq T_{\mathrm{eff}} \leq 8500\,\mathrm{K}$) and 10 metallicity values ($-3.0 \leq [\mathrm{Fe/H}] \leq +1.5$).

The expanded raw dataset of 3,667 wavelengths per star was standardized and compressed via PCA into 10 principal components. The Bayesian classifier was retrained using a 75/25 train-test split to evaluate algorithmic accuracy on blind test stars.

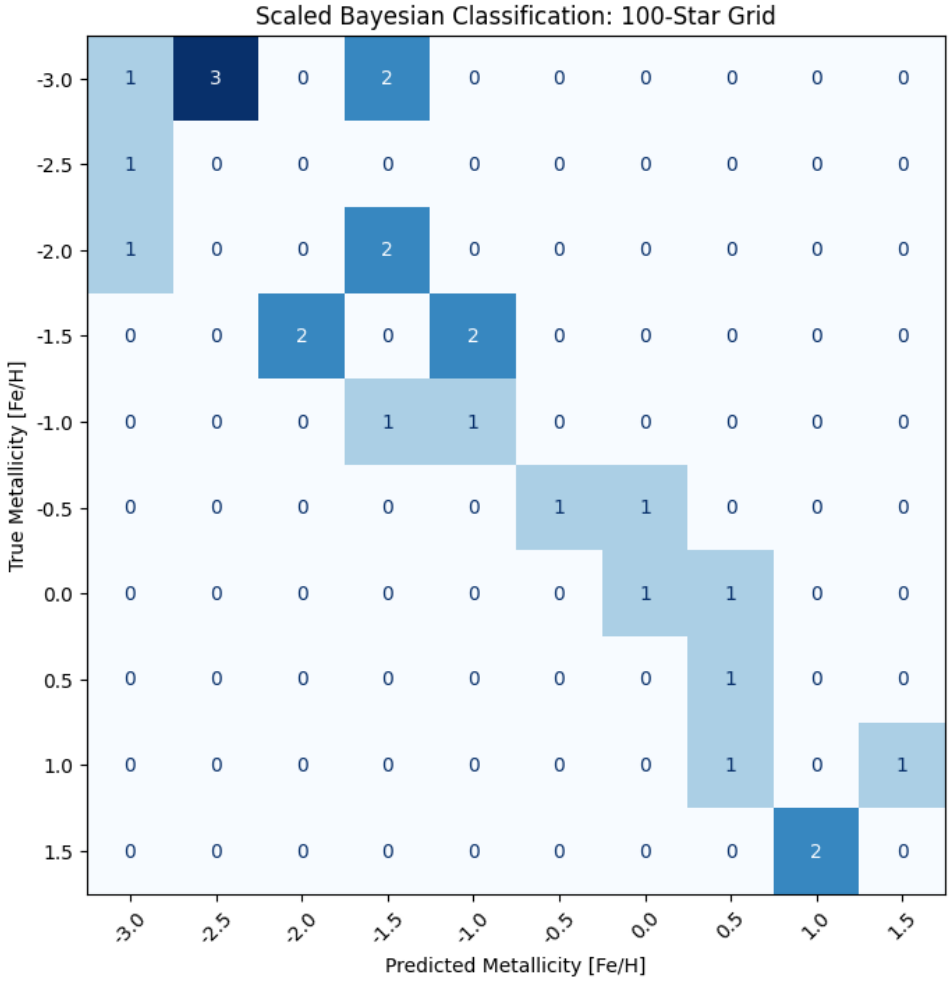


Figure 1: The 10×10 Confusion Matrix representing the Phase 2 scaled Bayesian classification. Predictions closely track the main diagonal, with variations restricted exclusively to physically adjacent metallicity classes.

As illustrated in Figure 1, the Phase 2 pipeline successfully resolved the statistical constraints of the prototype. The confusion matrix demonstrates a strong diagonal trend. Crucially, the remaining classification variances are restricted to immediately adjacent metallicity categories (e.g., a true −1.5 [Fe/H] predicted as −1.0 or −2.0). This confirms the Bayesian model successfully learned the underlying atomic absorption physics rather than memorizing data noise.

4 Limitations

- Noise-free synthetic data and a single fixed $\log g$ (and α): results are an upper bound for real data.

- Every test star lies on the training grid, so interpolation between grid values is untested.
- Decimated rather than resolution-matched spectra; no continuum normalisation of the kind used on observed spectra.
- A 0.3 dex RMSE is still large compared with typical survey precision, and the metal-poor end ($[Fe/H] < -2$) is hardest because lines are weak.
- Median normalisation is not a proper continuum fit.

5 Conclusion

As a first test, PCA plus Gaussian Naive Bayes on raw PHOENIX fluxes is a weak metallicity estimator: 20% exact-bin accuracy, 88% within one bin. The main lesson is that temperature dominates the variance, so normalisation and regression matter more than the choice of classifier. This is a transparent baseline, and the code is ready for the realistic tests above.

References

[1] Husser, T.-O., et al. 2013, A&A, 553, A6 (PHOENIX spectral library).

[2] Pedregosa, F., et al. 2011, JMLR, 12, 2825 (scikit-learn).